LUMEN / RATE INTELLIGENCE

Independent filing review memo

Juniper Lantern Mutual Insurance Company (fictional)
Connecticut · Personal Auto · Effective 2027-01-01

Review b94b748c6d534285b830e7fc71b596e7 · 2026-10-25T00:14:23.753556+00:00
Saved real AI review · Status: complete

> **Human review required.** Synthetic data only. Connecticut is context; state-specific regulatory requirements have not been verified or implemented. This memo is analyst decision support and makes no legal-compliance or submission-approval determination.

Executive assessment

8 findings returned; 5 high severity. Both specialized agent stages completed.

Rate Analysis · complete

The exposure-weighted rate-level change is 6.710625, while fixed-exposure premium impact is 7.84443. These describe different measures. Current-premium weighting is unavailable because current premiums are inconsistent. Deterministic findings identify mismatches in the claimed overall change, segment changes, and a supplied premium.

Documentation Review · complete

The submitted material contains unsupported factor-change rationale, an unsupported approval assertion, and a narrative claim contradicted by the calculated segment evidence. These are documentation findings, not a legal-compliance determination.

Portfolio impact

Measure	Calculated	Basis / limitation
Exposure-weighted average rate change	6.7106 %	rate_level
Fixed-exposure portfolio premium impact	7.8444 %	premium_impact
Current modeled portfolio premium	6,493,567.50 USD	premium_total
Proposed modeled portfolio premium	7,002,950.86 USD	premium_total
Current-premium-weighted rate change	Unavailable	inconsistent_current_premiums

Submitted overall claim: 1.000000% on fixed_exposure_portfolio basis.

Findings and recommended actions

HIGH · DETERMINISTIC · SEGMENT_CHANGE_MISMATCH:PA003

Segment Change Mismatch

Reported segment change differs from the independently calculated rate change.

Next action: Reconcile the reported percentage against the proposed base rate and combined factor.

Evidence: row:PA003, calc:PA003

HIGH · DETERMINISTIC · CURRENT_PREMIUM_MISMATCH:PA011

Current Premium Mismatch

Supplied current premium is inconsistent with exposure times the modeled annual rate.

Next action: Correct the premium or clarify the premium basis before using premium weights.

Evidence: premium:PA011

HIGH · DETERMINISTIC · SEGMENT_CHANGE_MISMATCH:PA020

Segment Change Mismatch

Reported segment change differs from the independently calculated rate change.

Next action: Reconcile the reported percentage against the proposed base rate and combined factor.

Evidence: row:PA020, calc:PA020

MEDIUM · DETERMINISTIC · LARGE_RATE_CHANGE:PA020

Large Rate Change

Absolute segment rate change exceeds the configured screening threshold; this is not a regulatory limit.

Next action: Confirm the factor revision and explain the large change to the analyst.

Evidence: calc:PA020

HIGH · DETERMINISTIC · OVERALL_CHANGE_MISMATCH:PORTFOLIO

Overall Change Mismatch

Claimed overall change differs from the calculated result on the explicitly stated basis.

Next action: Reconcile the filing summary to the calculated portfolio result.

Evidence: metadata:filing, metric:fixed_exposure_portfolio

MEDIUM · AI · AI-DOCUMENTATION_REVIEW-1

Factor-change support not provided

The narrative says combined factor revisions are described in DOC-FACTORS, but the inventory marks that document not provided. The filed explanation therefore lacks the cited supporting rationale.

Next action: Provide the factor-change explanation referenced by the narrative.

Evidence: narrative:submitted, doc:DOC-FACTORS

HIGH · AI · AI-DOCUMENTATION_REVIEW-2

Unsupported approval assertion

The internal note asserts automatic approval but supplies no authoritative source, effective date, statute, DOI instruction, or approval record. This assertion is unsubstantiated in the submitted materials and is not verified here.

Next action: Remove the assertion or provide authoritative supporting material for review.

Evidence: doc:DOC-APPROVAL

MEDIUM · AI · AI-DOCUMENTATION_REVIEW-3

Narrative rate-bound claim contradicted

The narrative claims every segment change stays within its stated bound. The bound check contradicts that claim and identifies PA020 as the violating segment; its calculation evidence shows the segment change.

Next action: Correct the narrative claim and reconcile the explanation for the affected segment.

Evidence: narrative:submitted, bound:57ba8e5880534514, calc:PA020

Segment exhibit

Segment	Current $	Proposed $	Change $ / exposure	Change %	Evidence
PA001	408.00	419.83	11.83	2.9000%	calc:PA001
PA002	480.00	504.00	24.00	5.0000%	calc:PA002
PA003	576.00	616.90	40.90	7.1000%	calc:PA003
PA004	768.00	838.66	70.66	9.2000%	calc:PA004
PA005	484.50	493.56	9.06	1.8710%	calc:PA005
PA006	570.00	592.52	22.52	3.9500%	calc:PA006
PA007	684.00	725.24	41.24	6.0290%	calc:PA007

Segment	Current $	Proposed $	Change $ / exposure	Change %	Evidence
PA008	912.00	985.94	73.94	8.1080%	calc:PA008
PA009	535.50	556.54	21.04	3.9290%	calc:PA009
PA010	630.00	668.12	38.12	6.0500%	calc:PA010
PA011	756.00	817.77	61.77	8.1710%	calc:PA011
PA012	1,008.00	1,111.74	103.74	10.2920%	calc:PA012
PA013	586.50	603.51	17.01	2.9000%	calc:PA013
PA014	690.00	724.50	34.50	5.0000%	calc:PA014
PA015	828.00	886.79	58.79	7.1000%	calc:PA015
PA016	1,104.00	1,205.57	101.57	9.2000%	calc:PA016
PA017	663.00	695.87	32.87	4.9580%	calc:PA017
PA018	780.00	835.38	55.38	7.1000%	calc:PA018
PA019	936.00	1,022.51	86.51	9.2420%	calc:PA019
PA020	1,248.00	1,737.59	489.59	39.2300%	calc:PA020
PA021	739.50	745.73	6.23	0.8420%	calc:PA021
PA022	870.00	895.23	25.23	2.9000%	calc:PA022
PA023	1,044.00	1,095.76	51.76	4.9580%	calc:PA023
PA024	1,392.00	1,489.66	97.66	7.0160%	calc:PA024

Methods and assumptions

Current rate = current base × current factor; proposed rate = proposed base × proposed factor. Segment change d = 100 × (proposed − current) / current. Exposure-weighted average = Σ(E × d) / ΣE. Fixed-exposure premium impact = 100 × [Σ(E × proposed) / Σ(E × current) − 1]. Current-premium-weighted change = Σ(P × d) / ΣP. Weights are never substituted. Missing, incomplete, negative or all-zero weights make the corresponding measure unavailable. Inconsistent supplied premiums invalidate premium weighting.

USD per annual vehicle exposure; fixed coverage and portfolio mix. Excludes taxes, fees, minimum premiums, policy rounding, retention and demand. Percentages are displayed rounded; source Decimal values remain in the JSON export. Large-change screening is a review heuristic, not a legal threshold.


```
{
  "change_tolerance_pp": "0.01",
  "premium_tolerance_usd": "0.01",
  "large_change_threshold_pct": "15"
}
```


Submitted documentation · untrusted source text

```
# Synthetic Personal Auto filing narrative

Juniper Lantern Mutual Insurance Company is fictional. All territories, segments,
premiums, and selections in this exercise are synthetic.

The proposed fixed-exposure portfolio premium impact is 1.000000%. This is the ratio of
total proposed modeled premium to total current modeled premium, minus one. The
exposure-weighted average segment change is a different measure.

The proposal increases the base rate by 5% and revises combined factors as described in
DOC-FACTORS. No segment changes by more than 15% in absolute terms. Current premiums
follow DOC-PREMIUM.

Connecticut provides state context only. This package makes no claim of DOI approval or
legal compliance. All materials are illustrative and are not suitable for regulatory
submission.

The filing team relies on the automatic-approval assertion in DOC-APPROVAL.
```

DOC-RATES · Synthetic rate exhibit

provided

```
The accompanying rates.csv and rates.xlsx show the same 24 disjoint rating segments.
Annual base rate changes from USD 600 to USD 630. Combined factors multiply the base
rate. Reported percentages use 5.0 to mean 5 percent.
```

DOC-FACTORS · Factor change explanation

not_provided

```

```

DOC-PREMIUM · Exposure and premium basis

provided

```
Exposures are annual vehicle-years in disjoint segments of a fixed portfolio. Current premium in USD equals exposures times the current annual modeled rate. Coverage, limits, deductibles and portfolio composition are held fixed. Fees, taxes, minimum premiums and individual-policy rounding are excluded. This is a simplified exercise, not a complete carrier rating algorithm.
```

DOC-APPROVAL · Internal approval note

provided

```
An internal note asserts that Connecticut automatically approves all Personal Auto increases below 20 percent. No authoritative source, effective date, statute, DOI instruction, or approval record is supplied to support this assertion.
```

Evidence appendix

bound:57ba8e5880534514

calc:PA003

calc:PA020

doc:DOC-APPROVAL

doc:DOC-FACTORS

metadata:filing

metric:fixed_exposure_portfolio

narrative:submitted

premium:PA011

row:PA003

row:PA020

Input fingerprints

- rates.csv: `461e901a50f937a6554bc99e8009d4325670ce8b0595c530199348dbee2f6f51`
- filing.json: `e6d7d0ab0b7b6b1bca95074dbf6bbba7f0a70fb8943d75f9cadf23e1e3fc2f96`
- narrative.md: `0730b9d54531bfe9efd22b5cb8618cf7d57cfec0f88ba475f151a8082de28a8c`

Execution and usage

Model: gpt-6-luna. 6 model requests; 17.59 seconds of measured model latency. Input: 31844; cached input (subset): 21745; output: 2341; total: 34185 tokens. Exact billing is unavailable. The JSON export includes the full sanitized execution ledger.